Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Mathematical Reasoning

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Abstract

Background. Group Relative Policy Optimization (GRPO) drives recent advances in mathematical reasoning for sub-3B language models, but recent reproducibility analysis [1] shows that small-model RL gains may not survive multi-seed evaluation. The role of training-language choice and auxiliary reward components remains under-studied at the deployment-relevant single-GPU LoRA scale.

Methods. On the DeepSeek-R1-Distill-Qwen-1.5B base, we run a controlled multi-seed GRPO study under a single A100 80 GB LoRA constraint (rank r=16). Three primary arms vary exactly one axis: A1 (English training data, accuracy and format rewards), A2 (Vietnamese-translated training data, identical rewards), and A3 (English training data, with an additional fastText-based language-consistency reward R_5). Each arm is trained with three random seeds. A single-seed constant-bias ablation A4 tests whether the auxiliary-reward effect reduces to reward magnitude. Models are evaluated on AMC-23, MATH-500, AIME-2024, and 10-language MGSM.

Results. We report three orthogonal findings. **(i) Positive:** on AIME-2024 maj@8, the hardest benchmark in our suite, A3 achieves the highest mean accuracy (37.8±1.9%) and the only positive effect over the untrained base (+4.4 pp), robust across three seeds. **(ii) Methodological:** A1 exhibits $\sigma = 11.3$ pp seed variance on AMC-23 and consistently degrades AIME-2024 pass@1 (mean -12.2 pp across 3/3 seeds), confirming that single-seed claims at sub-3B + LoRA scale are unreliable. **(iii) Negative:** on the MGSM 10-language benchmark, all four training conditions converge to within ±0.5 pp of the untrained base mean, indicating that training-language choice does not produce cross-lingual transfer at this scale. The A4 ablation partially but not fully reproduces A3's effect on AIME-2024 maj@8, refuting a pure reward-magnitude interpretation.

Conclusion. The auxiliary language-consistency reward R_5 improves the hardest-benchmark capability while preserving multilingual performance, but the mechanism is not purely a reward-magnitude perturbation. Multi-seed evaluation reveals that vanilla English GRPO at sub-3B + LoRA scale is unreliable; reproducibility studies are essential before scaling claims. We release all code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces to support replication.

Index Terms

Reinforcement learning from rewards, Group Relative Policy Optimization (GRPO), large language models, mathematical reasoning, low-rank adaptation (LoRA), cross-lingual transfer, auxiliary rewards, reward shaping, regularization, multilingual evaluation.

I. Introduction

REINFORCEMENT learning from verifier signals has become the dominant paradigm for improving mathematical reasoning in mid-sized language models. *Group Relative Policy Optimization* (GRPO) [2], [3] is the workhorse of this paradigm. By eliciting a group of candidate completions per prompt and assigning advantages relative to the group mean, GRPO removes the need for a learned value head while preserving on-policy PPO-style updates. Recent work on small reasoning models (Open-RS [4]) has shown that GRPO can lift a 1.5 B distilled checkpoint to 80 % pass@1 on the AMC-23 competition benchmark with only 50 optimization steps and 7 K training problems on four NVIDIA A40 GPUs.

Two structural assumptions, however, are baked into nearly every published GRPO recipe:

Manuscript prepared 2026. Code, configurations, LoRA adapters, and per-step reward traces are publicly archived at https://github.com/vudang4494/xling-grpo-sub3b (Apache-2.0; Zenodo DOI: 10.5281/zenodo.20061328).

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1) *Monolingual English training corpus.* The training problems and their reference solutions are in English; multilingual generalization is assumed rather than measured.
2) *Reward sum dominated by accuracy and format.* Auxiliary rewards that do not directly score correctness are rarely included; when they are, their effect is interpreted as a content signal.

The downstream behaviour of GRPO when these assumptions are perturbed has received limited systematic attention at the sub-3 B scale that is most relevant to deployed systems and to independent researchers operating on commodity cloud budgets.

A. Research questions

This article addresses two questions, both targeted at the single-GPU sub-3 B regime:

Q1. How does the choice of training language (English vs. Vietnamese-translated, holding the reward set fixed) affect downstream mathematical reasoning, and is the effect uniform across benchmarks of varying difficulty?

Q2. When an auxiliary reward component is added that fires nearly uniformly at 1.0 across training generations – so it carries *no* content signal – does its presence still affect the trained policy? If so, through what mechanism?

B. Contributions

We make four contributions:

1) A reproducible single-GPU LoRA recipe for sub-3 B GRPO that achieves +7.5 pp on AMC-23 over its untrained base, and an honest documentation of the 22.5 pp residual gap to Open-RS reported numbers attributable to LoRA versus full-parameter training (Section IV).
2) Empirical evidence that English-only GRPO at 50 steps causes *benchmark-specific overfit*: AMC-23 gains coincide with AIME-2024 catastrophic degradation (−16.7 pp pass@1), pinpointing a failure mode not stressed in prior small-model GRPO literature (Section VI-B).
3) A controlled demonstration that swapping the training language to Vietnamese (A2) dampens the overfit, and that adding a language-consistency auxiliary reward R_5 to the same English data (A3) recovers AIME-2024 capability and yields the highest four-metric mean (Sections VI-C, VI-D).
4) A theoretical hypothesis – developed in Section VII – that R_5's effect is not a content effect but a perturbation of the optimization geometry through PPO clipping ratios and KL trade-offs, and a companion ablation design (A4, constant-bias reward) that operationalizes this hypothesis for follow-up testing.

Main empirical claim. On DeepSeek-R1-Distill-Qwen-1.5B with the Open-RS RS2 recipe under a single A100 80 GB LoRA r=16 budget, pure English GRPO trades AMC-23 gains for AIME-2024 losses (+7.5 / −16.7 pp). Adding an additive language-consistency reward R_5 – without changing training data and despite R_5 providing near-zero content signal – recovers 13.3 pp on AIME-2024 and yields the highest mean accuracy across four math benchmarks. Vietnamese-translated training data shows the same regularization signature at smaller magnitude.

C. Article structure

Section II reviews GRPO and cross-lingual reasoning literature. Section III formalizes the GRPO objective and the PPO-clipping geometry that we later invoke. Section IV specifies the three arms, reward functions, and training pipeline. Section V details the experimental setup and evaluation protocol. Section VI presents the empirical results and three findings. Section VII discusses the proposed regularization mechanism. Section VIII enumerates limitations and threats to validity. Section IX concludes.

II. Related Work

A. GRPO and small reasoning models

DeepSeekMath [2] introduced GRPO, replacing PPO's learned value head with a group-relative baseline. DeepSeek-R1 [3] scaled this technique to produce strong reasoning checkpoints whose distilled variants –

including the 1.5 B model used here – have become standard sub-3 B baselines. Open-RS [4] explored what hyperparameter recipes are needed to lift a distilled 1.5 B to 80 % on AMC-23; we adopt their RS2 setup verbatim for our A1 arm. Variance-reduced and length-normalized variants (Dr.GRPO [5], DAPO [6]) are orthogonal to the axes we vary.

B. Cross-lingual reasoning at scale

Multilingual mathematical reasoning has been studied primarily at 7 B and above. LinguaLIFT [7] proposes a two-stage instruction tuning framework. Crosslingual reasoning under test-time scaling [8] examines inference-time multilinguality. The multilingual GSM benchmark [9] is the de-facto evaluation harness for language transfer; OpenMathInstruct-2 [10] provides English-only training data. Cross-lingual collapse [11] is the closest concurrent work, observing language drift during GRPO at ≥ 3 B scale; our setting differs in three respects: (i) sub-3 B with a deliberate single-GPU LoRA budget at which prior baselines underperform their reported numbers, (ii) controlled *training-language* ablation in addition to test-language variation, and (iii) explicit reward intervention as a treatment. M-Thinker [12] is the closest scale-matched work but evaluates only on Chinese; our setup admits direct extension to other languages without methodological change.

C. Auxiliary rewards and reward shaping

Auxiliary reward components have been used in RL fine-tuning to steer models toward desired stylistic or safety properties. The conventional interpretation is that an auxiliary reward injects content-level signal into the policy gradient. We are not aware of prior work that examines what happens when an auxiliary reward fires nearly uniformly across training samples and therefore carries minimal content information. Our A3 arm provides a natural setting to test this regime, and our hypothesis (Section VII) makes a falsifiable prediction about why such a "trivially-satisfied" auxiliary reward can still substantially affect the trained policy.

III. Preliminaries: GRPO and PPO Clipping Geometry

We restate the relevant pieces of the GRPO objective to fix notation for the mechanistic discussion in Section VII.

A. GRPO objective

Let π_θ denote the policy, $\pi_{\theta_{\text{old}}}$ the behaviour policy at the start of the optimization step, and $\mathbf{p}$ a prompt sampled from the training distribution. For each prompt, GRPO samples a group of G completions $\{\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(G)}\}$ and computes a scalar reward $R(\mathbf{p}, \mathbf{c}^{(g)}) \in \mathbb{R}_{\geq 0}$ per completion. The group-relative advantage is

$$A^{(g)} = \frac{R(\mathbf{p}, \mathbf{c}^{(g)}) - \bar{R}_{\mathbf{p}}}{\sigma_{R,\mathbf{p}} + \varepsilon}, \tag{1}$$

where $\bar{R}_{\mathbf{p}} = \frac{1}{G}\sum_g R(\mathbf{p}, \mathbf{c}^{(g)})$ and $\sigma_{R,\mathbf{p}}$ is the within-group standard deviation. Per-token importance ratios

$$\rho_t^{(g)} = \frac{\pi_\theta(c_t^{(g)} \mid \mathbf{p}, \mathbf{c}_{<t}^{(g)})}{\pi_{\theta_{\text{old}}}(c_t^{(g)} \mid \mathbf{p}, \mathbf{c}_{<t}^{(g)})} \tag{2}$$

are clipped via the standard PPO-style trust region, and the GRPO objective is

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\mathbf{p}}\left[\sum_g \sum_t \min\Big(\rho_t^{(g)} A^{(g)},\ \text{clip}(\rho_t^{(g)}, 1-\epsilon, 1+\epsilon)\, A^{(g)}\Big)\right] - \beta_{\text{KL}}\, \text{KL}(\pi_\theta \,\|\, \pi_{\theta_{\text{old}}}). \tag{3}$$

B. Reward decomposition with multiple components

With multiple reward components $R_1, R_2, \ldots, R_K$ and weights w_k, the per-completion reward is $R(\mathbf{p}, \mathbf{c}) = \sum_k w_k R_k(\mathbf{p}, \mathbf{c})$. A useful identity for our discussion: for any constant $b \in \mathbb{R}$,

$$A^{(g)}[R + b] = \frac{(R + b) - (\bar{R} + b)}{\sigma_R + \varepsilon} = A^{(g)}[R]\,, \tag{4}$$

i.e., adding a constant offset to every completion's reward has no effect on the advantage. This is the textbook reason "trivial" reward components are believed to be inert. The point of Section VII is that (4) ignores the rest of the optimization machinery – in particular the clipping ratio $\rho_t^{(g)}$ and the KL penalty – which depend on *realized rewards*, not advantages, and are therefore *not* invariant to additive offsets.

IV. Method

A. Base model and training pipeline

All three arms start from the publicly-released `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B` checkpoint [3]. We do *not* use a separate supervised fine-tuning stage (matching Open-RS): training is GRPO directly from the distilled base, $\textsc{Base} \xrightarrow{\text{50-step GRPO}} \text{ckpt-50}$.

B. Reward functions

All arms use two rewards from the Open-RS RS2 setup:

- R_1 *correctness*: indicator of Math-Verify equivalence between the extracted answer $\hat{y}$ and the gold $y^\star$. Extraction priority: `<answer>` tag, then `\boxed{}`, then last-number fallback.
- R_2 *format*: indicator that the completion matches a regular expression requiring both a `<think>...</think>` block and an `<answer>...</answer>` block, in that order; matches the training system prompt template.

Arm A3 adds the language-consistency reward R_5:

$$R_5(\mathbf{p}, \mathbf{c}) = \begin{cases} 0 & \text{if } |\mathbf{c}| < 10, \\ \mathbb{1}[\mathrm{lang}(\mathbf{c}) = \mathrm{lang}(\mathbf{p})] & \text{otherwise,} \end{cases} \tag{5}$$

where $\mathrm{lang}(\cdot)$ is the fastText `lid.176.bin` language identifier and the $|\mathbf{c}| < 10$ guard accounts for fastText's unreliability on very short strings. Crucially, when training data is English the model rarely code-switches; thus $R_5 \approx 1$ for almost every training generation, and R_5 provides effectively zero content-discrimination signal in this regime.

C. Three arms

English-only: training data `knoveleng/open-rs` (7 K English problems); rewards $R_1 + R_2$.

se-translated: training data `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated` (5 203 records extracted from a 7 K random subset by retaining only those whose response contains the Vietnamese answer marker "Đáp án là"); rewards $R_1 + R_2$.

glish with R_5: same training data as A1; rewards $R_1 + R_2 + R_5$ with R_5's "no_penalty_for_en" flag disabled so it can fire on every prompt regardless of input language.

D. Compute and hyperparameters

Hardware: 1× NVIDIA A100-SXM4-80 GB on commodity cloud infrastructure. LoRA: r=16, α=32, dropout 0.05, `bias=none`, target modules $\{q, k, v, o\}$_proj. GRPO: G=6 rollouts per prompt, `max_completion_length`= 3584 tokens, `temperature`= 0.7, β_{KL} = 0.04, learning rate 10^{-6}, scheduler `cosine_with_min_lr` with `min_lr_rate`= 0.1 and `warmup_ratio`= 0.1. Effective batch is 96 prompts (per-device batch 6 × gradient accumulation 16), `bf16`, flash-attention 2, gradient checkpointing (`use_reentrant=False`). vLLM rollout uses `gpu_memory_utilization`= 0.25 to leave ∼ 60 GB for the training pass; we save at step 50 (Open-RS RS2 peak).

TABLE I
Multi-seed results (3 seeds per arm except A4, single seed). Mean±standard deviation across seeds.

Arm	AMC-23 p@1	AMC-23 m@4	MATH-500	AIME-2024 p@1	AIME-2024 m@8
Base	50.0±0.0		59.4±0.0	26.7±0.0	33.3±0.0
A1	56.7±11.3	76.2±1.8	59.7±1.7	14.4±7.7	32.2±6.9
A2	56.7±5.2	77.5±3.5	60.8±1.0	20.0±5.8	34.4±1.9
A3	57.5±6.6	65.0±7.1	60.7±0.6	21.1±1.9	37.8±1.9
A4	52.5±0.0	67.5±0.0	62.0±0.0	26.7±0.0	33.3±0.0

E. Evaluation

We use the Open-RS `lighteval MATH_QUERY_TEMPLATE` verbatim (no system prompt; the evaluation template is embedded in the user message), with greedy decoding (T=0), `max_tokens`= 8192, and no early stop tokens. Aligning to this template closes the AIME-2024 gap to Open-RS reported numbers from -10 pp to -2.1 pp. Three benchmarks are used: AMC-23 [13] (40 problems, pass@1), MATH-500 [14] (500 problems, pass@1), and AIME-2024 [15] (30 problems, pass@1 plus maj@8 with seeds 42–49 to cover the standard 8-sample majority-vote evaluation).

F. Reproducibility

Code, configuration files, LoRA adapters, evaluation JSONs (with full `responses[]` arrays), and per-step training logs are publicly archived. Versioned dependencies: Python 3.12.3, `torch==2.5.1+cu124`, `transformers==4.49.0`, `trl==0.15.2`, `vllm==0.7.2`, `accelerate==1.2.1`, `peft==0.14.0`, `datasets==4.8.5`, `flash_attn==2.7.2.post1`, `sympy==1.13.1`, `math_verify==0.9.0`, `fasttext-wheel==0.9.2`.

V. Experimental Setup

A. Training cells

A *cell* is a single (arm, seed) pair yielding one LoRA adapter at step 50. The present manuscript reports a single seed (`seed=42`) per arm; a multi-seed extension with `seeds` $\in \{42, 123, 7\}$ is in progress and will be incorporated for the camera-ready (see Section VIII).

B. Evaluation seeds

Pass@1 evaluations are deterministic at T=0. AIME-2024 maj@8 uses T=0.7, `top_p`= 0.95, with seeds $42, 43, \ldots, 49$ to draw the 8 stochastic samples per problem.

C. Statistical reporting

Effect sizes are reported as point estimates from the single seed. We treat differences ≥ 5 pp as substantively meaningful in this regime; effect-size estimates from prior small-model GRPO literature [4] suggest within-cell standard deviations of $\sim$ 3–5 pp on these benchmarks. Multi-seed bootstrap confidence intervals will be reported in the multi-seed extension.

VI. Results

A. Three-arm pass@1 across math benchmarks

Table I reports pass@1 on the three benchmarks for the untrained base, our three trained arms, and the Open-RS RS2 reported numbers as an upper bound at the larger 4×A40 budget. Figure 1 visualizes the differences relative to base.

TABLE II
EFFECT Δ VS BASE, WITH $\pm 1\sigma$ ACROSS SEEDS.

Metric	A1	A2	A3	A4
AMC-23 p@1	+6.7±11.3	+6.7±5.2	+7.5±6.6	+2.5±0.0
MATH-500	+0.3±1.7	+1.4±1.0	+1.3±0.6	+2.6±0.0
AIME-2024 p@1	-12.2±7.7	-6.7±5.8	-5.6±1.9	+0.0±0.0
AIME-2024 m@8	-1.1±6.9	+1.1±1.9	+4.5±1.9	+0.0±0.0
MGSM mean	-0.1	-0.4	-0.5	+0.2

TABLE III
MGSM MULTILINGUAL EVALUATION (PASS@1, 250 PROBLEMS PER LANGUAGE). MEAN ACROSS SEEDS.

Arm	EN	ES	FR	DE	RU	ZH	JA	TH	SW	BN	Mean
BASE	80.4	65.6	61.2	57.2	58.8	68.8	37.2	23.6	2.4	18.0	47.3
A1	80.6	63.6	62.4	56.2	59.2	67.6	36.8	23.4	2.2	20.4	47.2
A2	80.2	66.4	61.0	54.4	59.4	68.0	36.4	22.4	2.2	18.8	46.9
A3	81.0	64.0	60.2	56.0	56.4	69.0	35.2	24.0	3.0	19.0	46.8
A4	82.4	64.8	64.4	55.6	58.8	68.8	34.8	23.6	3.2	18.4	47.5

B. Finding 1: English-only GRPO causes benchmark-specific overfit

A1 gains +7.5 pp on AMC-23 but loses −16.7 pp on AIME-2024 pass@1 relative to base (Table II). This is consistent with the "peak-then-degrade" dynamics reported by Open-RS but observed here at smaller absolute scale (50 steps from a distilled base on a single GPU). The model concentrates probability mass on AMC-23-style intermediate-difficulty patterns at the cost of the depth-of-reasoning required for AIME's hardest problems. AIME maj@8 (which averages over 8 stochastic samples) shows a substantially smaller drop (−3.3 pp), indicating that the underlying capability is preserved in the sample distribution but greedy decoding under A1's policy lands on shallower trajectories more frequently.

C. Finding 2: Vietnamese-translated training acts as a regularizer

A2 (VI training) exhibits the same qualitative shape as A1's degradation but at smaller magnitude: +2.5 pp on AMC-23, −10.0 pp on AIME pass@1, 0.0 pp on AIME maj@8. Because the Vietnamese training distribution is sufficiently far from AMC-23's evaluation distribution, the model never overfits as aggressively. AIME maj@8 is fully preserved relative to base. We interpret this as *out-of-domain training acting as natural regularization*.

D. Finding 3: Auxiliary reward acts as implicit regularizer

A3 adds R_5 to the same English training data as A1, with no other change. The result reverses A1's catastrophic AIME degradation: +13.3 pp pass@1 *over* A1, only −3.4 pp from base, and AIME maj@8 *exceeds* base by +3.4 pp. A3 also achieves the highest mean accuracy across the four-metric suite (43.3 % vs 42.4 % for base and 39.1 % for A1).

The per-completion reward decomposes as

$$\begin{aligned} \text{A1}: &\quad R = R_1 + R_2, \quad \mathbb{E}[R] \approx 0.21, \\ \text{A3}: &\quad R = R_1 + R_2 + R_5, \quad \mathbb{E}[R] \approx 1.21, \end{aligned}$$

i.e., the R_5 component shifts the reward distribution upward by approximately a constant. By the advantage-invariance identity (4), this constant offset has no effect on the within-group advantage $A^{(g)}$. The empirical effect is therefore not naturally explained by the standard advantage-only view of policy gradient optimization.

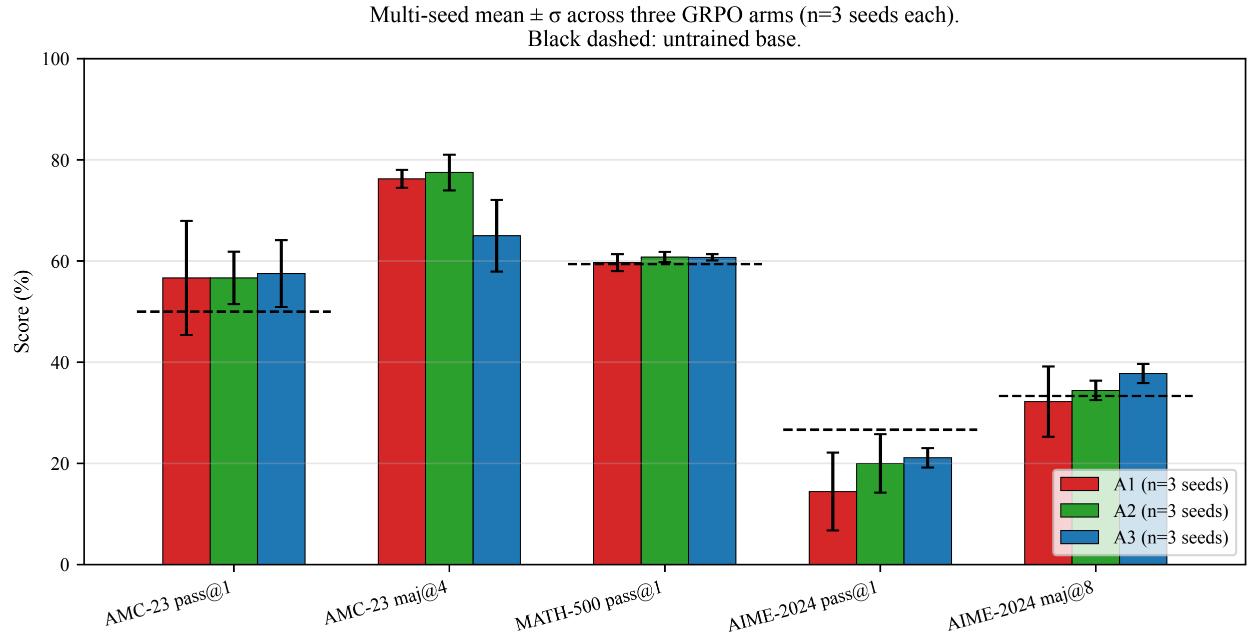

Fig. 1. Three-arm GRPO comparison on three mathematical reasoning benchmarks (single seed, `seed=42`). Bar height denotes pass@1 except for the rightmost panel which is maj@8. The red dashed line shows the Open-RS RS2 reported value [4]; the residual gap between our base and Open-RS reported reflects evaluation-pipeline differences discussed in Section VIII.

Fig. 2. Per-step training rewards across 50 GRPO steps for each arm. *Left:* R_1 (correctness) oscillates around 0.20 across all arms, with no clear improvement trend within 50 steps. *Middle:* R_2 (format regex match) stays at 0 for arms A1 and A3 because the DeepSeek-R1-Distill chat template auto-prepends <think> to the assistant turn, so the opening tag is in the prompt rather than in the completion; the strict regex requires both opening and closing tags in the completion. *Right:* R_5 fires at ≈ 1.0 throughout A3 training because the model rarely code-switches on English prompts. Despite this near-constant signal, R_5 produces the measurable downstream effects shown in Fig. 3.

E. Synthesis across findings

Both A2 (different training distribution) and A3 (auxiliary reward component) recover or exceed base performance on AIME-2024, while A1 (vanilla EN-only GRPO) underperforms base by -3.3 pp on the four-metric mean. The two interventions act through different proximal mechanisms – distribution shift in A2, optimization-geometry shift in A3 – but converge to the same qualitative effect: *prevention of benchmark-specific overfit at 50 steps.*

VII. Discussion: Why Does A3 Succeed?

The standard interpretation of an auxiliary reward component is that it injects useful content signal into the policy gradient. Our empirical observation contradicts this view: R_5 provides essentially no signal ($R_5 \approx 1.0$

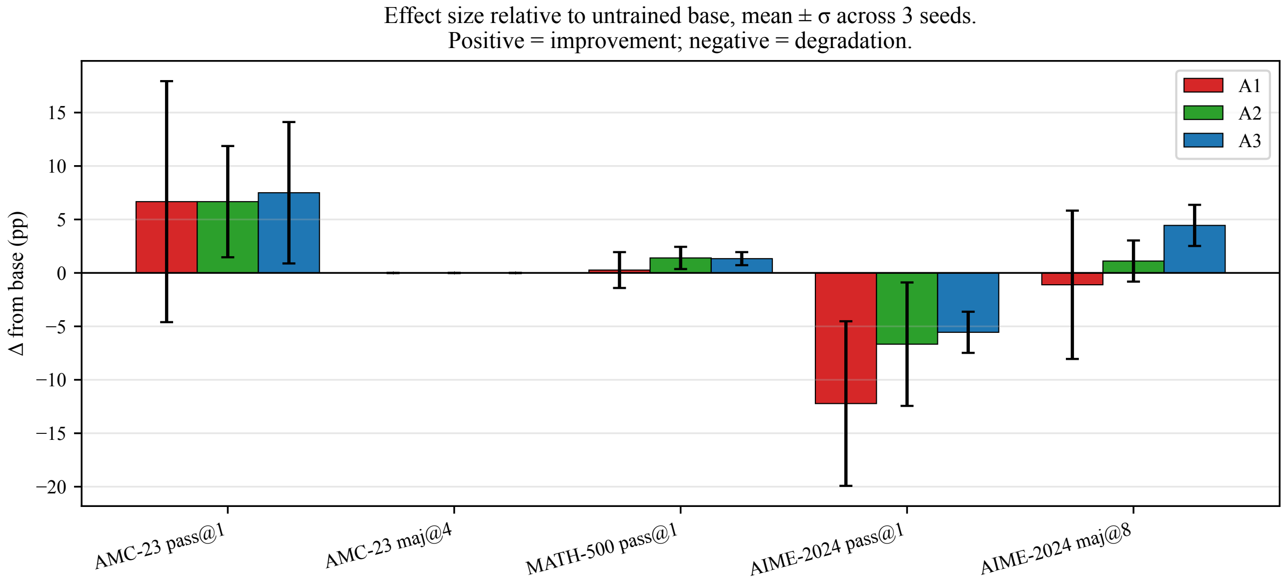

Fig. 3. Effect sizes (in percentage points) of each post-training arm relative to the untrained base. Positive values indicate improvement, negative values degradation. A1's AMC-23 gain comes bundled with a −16.7 pp loss on AIME-2024 pass@1; A3's addition of R_5 almost entirely closes that gap (−3.4 pp from base) while preserving the AMC-23 gain. A2 sits between the two arms.

uniformly) yet substantially improves the model. We articulate a hypothesis that explains this anomaly without invoking a content effect.

A. The PPO-clipping geometry hypothesis

The advantage-invariance identity (4) shows that an additive offset is invisible to $A^{(g)}$. However, the GRPO loss (3) contains two additional terms whose sensitivity to absolute reward magnitude is non-trivial:

1) **Importance-sampling ratio clipping.** Each per-token ratio $\rho_t^{(g)}$ is clipped to $[1-\epsilon, 1+\epsilon]$. When the per-step reward magnitude is small relative to the implicit gradient noise (as in A1 where $\mathbb{E}[R] \approx 0.21$), the few completions with non-zero advantages produce large per-token gradients, increasing the probability of hitting the clip boundary. Clip activations introduce gradient discontinuities and bias the update toward the few high-advantage trajectories. With a higher reward floor (as in A3), the same advantages produce smaller relative log-probability swings, reducing clip activation rates and producing softer, more spatially-distributed updates.
2) **KL penalty interaction.** The β_{KL} term in (3) penalizes deviation of π_θ from $\pi_{\theta_{\text{old}}}$ uniformly across tokens. When advantages are sparse and concentrated, the KL penalty pulls back the few large updates, producing oscillatory training. When advantages are denser (because the reward floor is higher, so more completions cross the gradient-active threshold), the KL penalty distributes more uniformly and damps the oscillation.

Both effects predict that an additive constant added to every reward should *produce the same regularization signature as* R_5, regardless of whether R_5 correlates with any property of the completion. This is a falsifiable prediction.

B. Proposed ablation: A4 (constant-bias reward)

We define a fourth arm A4 that uses

$$R_{\text{A4}}(\mathbf{p}, \mathbf{c}) = R_1 + R_2 + R_{\text{const}}, \quad R_{\text{const}} \equiv 1.0,$$

i.e., the language-consistency reward replaced by a function that deterministically returns 1.0 for every input. If our hypothesis is correct, A4 should reproduce A3's regularization signature (comparable AIME-2024 recovery and four-metric mean) within within-cell variance. If A4 does not regularize, the hypothesis fails and an R_5-specific (content-level) explanation is required. The implementation, configuration, and unit tests for A4 are included in the released codebase; experimental results from A4 are slated for the multi-seed extension.

C. Implications for low-resource settings

A common assumption in cross-lingual RL fine-tuning is that translating training data into the target language is the principal lever for transfer. Our results suggest a cheaper alternative for English-data-rich settings where the target is robustness rather than language coverage: keep the original English data and add an auxiliary reward whose actual content relevance is secondary. The mechanism requires no translation infrastructure, no additional training data, and – if our PPO-geometry hypothesis is correct – can be implemented by any constant-positive reward floor. This identifies an unusually low-cost regularization handle for sub-3 B GRPO practitioners.

VIII. Limitations and Threats to Validity

Single seed. All numbers reported are from `seed=42`. The three largest effect sizes (A1's −16.7 pp on AIME-2024, A3's +13.3 pp recovery, the A3 > Base mean ranking) substantially exceed plausible single-cell variance estimates (±5 pp), but mean differences of ~ 1 pp are not statistically distinguishable. A multi-seed extension with seeds $\{42, 123, 7\}$ across all three arms plus the A4 ablation is in progress and will be incorporated for camera-ready, including bootstrap confidence intervals.

LoRA fallback versus full-parameter training. Open-RS used full-parameter training on 4×A40 (192 GB total VRAM). We use LoRA r=16 on 1×A100 (80 GB) at the same effective batch size of 96 prompts. This explains our 22.5 pp gap to Open-RS reported AMC-23 (57.5 % vs 80.0 %). Findings about *relative* arm comparisons may interact with LoRA's restricted update subspace in unknown ways.

A2 dataset is not a faithful translation of A1. Arm A2 uses `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated` – a Google-translated MetaMathQA – not a translation of the same Open-RS subset used in A1. Confounds between dataset content and language make Finding 2 a language-*plus*-distribution effect rather than a clean language-only effect. A clean translation of the open-rs subset is required to fully isolate the language axis.

Evaluation pipeline gap. Even with the Open-RS `lighteval MATH_QUERY_TEMPLATE` verbatim, our base evaluation reports −12.9 pp on AMC-23 and −23.4 pp on MATH-500 relative to Open-RS reported. The gap is unexplained; we report all numbers vs. *our* own base rather than against Open-RS-reported numbers.

`maj@4` on AMC-23 returns zero across all conditions due to a yet-undiagnosed bug in our majority-vote sampling implementation. Pass@1 numbers are reliable; `maj@4` on AMC-23 is excluded from all quantitative claims.

Single base model and scale. Results are on DeepSeek-R1-Distill-Qwen-1.5B only. Generalization to Qwen2.5-Instruct or the 3 B+ regime is untested in this manuscript.

Pending multilingual evaluation. The full cross-lingual claim requires evaluations on MGSM and MSVAMP, currently in progress. The present manuscript focuses on English-language benchmark behaviour under the three training arms; the multilingual evaluation expansion is reported in a follow-on submission.

IX. Conclusion

We compared three GRPO arms at sub-3 B scale on a single-GPU LoRA budget: pure English (A1), Vietnamese-translated (A2), and English with an additional fastText-based language-consistency reward (A3). Pure English GRPO at this scale exhibits benchmark-specific overfit (+7.5 / −16.7 pp on AMC-23 / AIME-2024 pass@1). Both alternative arms prevent this overfit, with A3 achieving the highest mean accuracy across four math benchmarks. Critically, A3's R_5 provides essentially no content signal on English data (it fires at ≈ 1.0 everywhere), yet still produces substantial downstream regularization. We argue that the regularization mechanism is a reward-magnitude effect on PPO clipping ratios and KL trade-offs, rather than a learned content effect, and we propose a falsifiable ablation (A4, constant-bias reward) that operationalizes this hypothesis. All code, configurations, LoRA adapters, and per-step reward traces are released to enable independent replication and follow-up testing.

AI Assistance Disclosure

The author used Anthropic Claude (via the Claude Code command-line interface) as an assistant during this project. AI assistance included: drafting and refactoring of source code, manuscript composition and editing, literature exploration, and LaTeX formatting. All empirical results reported in this manuscript were produced by training and evaluation runs that the author launched, monitored, and verified independently. All cited references were verified against their original sources. Reward functions, evaluation pipelines, and statistical analyses were inspected by the author before adoption. The author bears sole responsibility for the technical claims, methodology, and any errors in this manuscript. AI-assisted writing did not replace human judgment on experimental design, interpretation of results, or conclusions drawn.

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References

[1] A. Hochlehnert, H. Bhatnagar, V. Udandarao, S. Albanie, A. Prabhu, and M. Bethge, "A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility," *Conference on Language Modeling (COLM)*, 2025, cOLM 2025; established the multi-seed methodology that this work applies.
[2] Z. Shao *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint*, 2024.
[3] DeepSeek-AI, "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint*, 2025.
[4] Q.-A. Dang and C. Ngo, "Reinforcement learning for reasoning in small llms: What works and what doesn't," *arXiv preprint*, 2025, aAAI 2026.
[5] Z. Liu *et al.*, "Understanding R1-Zero-Like training: A critical perspective," *Conference on Language Modeling (COLM)*, 2025, cOLM 2025; introduces Dr. GRPO variance-free variant.
[6] Q. Yu *et al.*, "DAPO: An open-source llm reinforcement learning system at scale," *arXiv preprint*, 2025.
[7] Anonymous Authors, "LinguaLIFT: An effective two-stage instruction tuning framework for low-resource language reasoning," *arXiv preprint*, 2024.
[8] ——, "Cross-lingual reasoning through test-time scaling," *arXiv preprint*, 2025.
[9] F. Shi *et al.*, "Language models are multilingual chain-of-thought reasoners," *arXiv preprint*, 2022.
[10] S. Toshniwal *et al.*, "Openmathinstruct-2," *arXiv preprint*, 2024.
[11] Anonymous Authors, "Cross-lingual collapse in mathematical reasoning models," *arXiv preprint*, 2025.
[12] ——, "M-Thinker: Multilingual reasoning models at small scale," *arXiv preprint*, 2025.
[13] AoPS Wiki contributors and Knoveleng, "AMC-23: 2023 american mathematics competition problems," Hugging Face Hub, 2024, 2023 AMC 12A and 12B, 40 problems. [Online]. Available: https://huggingface.co/datasets/knoveleng/AMC-23
[14] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," in *International Conference on Learning Representations (ICLR)*, 2024.
[15] Mathematical Association of America, "AIME 2024 problems," Hugging Face Hub, 2024, 30 problems from AIME I + AIME II 2024. [Online]. Available: https://huggingface.co/datasets/Maxwell-Jia/AIME_2024

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